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MILITARY SPECIFICATION
FILTER ELEMENT AND RELIEF VALVE, FLUID PRESSURE, HYDRAULIC (M-60 TANK SERIES)

This specification is approved for use by the U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance and manufacture requirements for a hydraulic filter element and relief valve, disposable type, 3 and 5 micron filtration, herein referred to as the assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 4820

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SPECIFICATIONS

MILITARY

MIL-H-46170	- Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic Hydrocarbon Base.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-810B	- Environmental Test Methods.
MIL-STD-1246	- Product Cleanliness Levels and Contamination Control Program.
MIL-STD-45662	- Calibration System Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Navy Publication and Printing Service Office, Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

12257235	- Filter Element and Relief Valve Assembly.
12270316	- Filter Element and Relief Valve Assembly.

(Copies of the Government drawings listed above as required by contractors in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article (preproduction). When specified (see 6.2), a preproduction sample of assemblies to be furnished under this specification shall be produced prior to the manufacture of the items, and shall be tested as specified herein for compliance of requirements listed below (see 4.4, 6.3).

3.2 Initial production. Unless otherwise specified (see 6.2), a sample of initial production assemblies (see 4.5) shall be subjected to inspection and tests specified herein to verify conformance to the requirements listed below. Approval of these assemblies by the activities concerned shall not relieve the contractor of the obligation to supply assemblies in production that are fully representative of those inspected in initial production testing. Any changes or deviations of the production assemblies from the initial production assemblies shall be subject to the approval of the contracting officer.

3.3 Materials. Material selection shall be approved by the contracting officer for compliance to the following criteria (see table VII, table VIII).

3.3.1 Metals. All metals used in the construction of the assembly shall be corrosion-resistant or shall be adequately protected to resist corrosion during the normal service life of the filter element.

3.3.2 Dissimilar metals. The use of dissimilar metals in contact with each other shall be avoided. Where contact between dissimilar metals is unavoidable, they shall be suitably protected against electrolytic corrosion.

3.3.3 Filter element material. The material of the filter element shall not change the composition of the hydraulic fluid MIL-H-46170, nor shall the material be altered enough in use or storage to affect the intended function of the filter element.

3.4 Design and construction. Assembly shall conform to Drawing 12257235 or 12270316 and other applicable drawings, specifications, and standards. Parts shall be interchangeable as to fit, form, and function with all other units bearing same part number. Assembly shall be interchangeable with accepted assemblies and with current mountings (see 4.6.2.2).

3.4.1 Relief valve. The assembly shall incorporate a permanently installed, nonadjustable, by-pass relief valve capable of performance as specified in 3.5.9 (see 4.7.2.9).

3.4.2 Flow direction. Flow through the element shall be from the outside in (see 4.7.1.1).

3.4.3 Dimensional constraints. Dimensional constraints of the assembly shall be as specified on Drawing 12257235 or 12270316, and drawings referenced therein (see table VII).

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3.4.4 Weight (see 4.7.1.2). The weight of the assembly shall not exceed 1.5 pounds (per Drawing 12257235) or 0.8 pounds (per Drawing 12270316).

3.5 Performance.

3.5.1 Bubble point. There shall be no bubbles visible when the filter element is pressurized to 0.1 inch of water less than the minimum bubble point determined during preproduction testing (see 4.7.2.1).

3.5.2 Filtration. The filter element shall remove 97 to 99 percent by weight of all contaminants as specified in table I (see 4.7.2.2).

TABLE I. Contaminants filtration.

Characteristic	Assembly		
	12257235	12270316-1	12270316-2
Filtration (microns)	5	3	3
Flow Rate (gpm)	10	12	3
Pressure Drop (psid)	6	12	10

3.5.3 Flow rate. The rated flow capacity of the filter element shall be as specified in table I (see 4.7.2.3).

3.5.4 Pressure drop. The pressure drop across the clean filter element shall not exceed the amount specified in table I at flow rate specified (see 4.7.2.4).

3.5.5 Dirt capacity. The filter element shall hold the specified amount of AC Fine Test Dust without exceeding the differential pressure as specified in table II at flow rate specified in 3.5.3 (see 4.7.2.5).

TABLE II. Dirt capacity.

Assembly	Incremental Additions of Contaminant (grams)	Minimum Capacity (grams)	Maximum Differential Pressure (psid)
12257235	2.5	20.0	100
12270316-1	1.2	4.8	45
12270316-2	0.6	2.6	45

3.5.6 Efficiency. The assembly shall remove 97 percent, by weight, of contaminant from a mixture of 2000 ml of clean hydraulic fluid MIL-H-46170 (less than 0.0004 grams of contaminant per 2000 ml of fluid), and a 5 ml slurry in accordance with table III (see 4.7.2.6).

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TABLE III. Slurry mixture.

Assembly	A/C Fine Test Dust (grams)	APM F-9 Beads (grams)
12257235	.15	.15
12270316-1	.25	.25
12270316-2	.15	.15

The efficiency shall be determined by the following formula:

$$\text{Efficiency} = \frac{A-(B-C)}{A} \times 100$$

Where: A=(Add value) = Amount of test contaminant passed through system when there is no filter element in the filter housing.

B=(contaminant value) = Amount of test contaminant passed through the test filter assembly with the filter element installed.

C=(blank value) = Amount of contaminant attributed to the test system and test filter assembly when no contaminant has been added. This amount shall be less than 0.0007 grams.

3.5.7 Collapse pressure. The filter element shall withstand the minimum differential pressure, at flow rate specified in table IV without inward structural failure or filter media failure. The differential pressure shall be obtained by adding AC Fine Test Dust to the test system while maintaining flow rate. The bubble point shall not be below the minimum established during preproduction or initial production testing (see 4.7.2.7).

TABLE IV. Collapse pressure.

Assembly	Differential Pressure (pounds per square inch) (psi)	Rated flow (gallons per minute) (gpm)
12257235	250 psi	10 gpm
12270316-1	150 psi	12 gpm
12270316-2	150 psi	13 gpm

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3.5.8 Flow fatigue. The filter element shall be capable of withstanding flow fatigue cycling specified in table V without evidence of structural failure. The bubble point shall not be less than the minimum value established during preproduction or initial production testing (see 4.7.2.8).

TABLE V. Flow fatigue.

Number cycles	Pressure Drop (pounds per square inch gage)	
	12257235	12270316-1-2
225,000	20	20
22,500	50	50
2,000	100	75
500	150	100

3.5.9 Relief valve performance.

3.5.9.1 Cracking pressure. The relief valve shall crack at the differential pressure specified in table VI. Cracking pressure is defined as that pressure at which leakage through the valve exceeds that specified in table VI (see 4.7.2.9.1).

TABLE VI. Relief valve

Assembly	Flow Capacity		Cracking		Reseat	
	Rate	Diff. Press.	Diff. Press.	Leakage (cubic centimeters per min.)	Diff. Press.	Leakage
12257235	15.5 gpm	190 psi	125 ± 25 psi	16 ccm	60 psi	8 ccm
12270316-1	18.0 gpm	120 psi	50 ± 5 psi	4 ccm	30 psi	2 ccm
12270316-2	4.5 gpm	120 psi	50 ± 5 psi	4 ccm	30 psi	2 ccm

3.5.9.2 Flow capacity. With the flow specified in table VI through the relief valve, the differential pressure across the relief valve shall not exceed the pressure specified in table VI (see 4.7.2.9.2).

3.5.9.3 Reseat pressure. The relief valve shall reseat at the differential pressure as specified in table VI. Reseat pressure is defined as that pressure at which leakage through the valve does not exceed the amount specified in table VI (see 4.7.2.9.3).

3.5.9.4 Endurance. The relief valve shall be capable of withstanding 2000 cycles of opening and closing the valve from zero to the flow capacity specified in table VI (see 4.7.2.9.4).

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3.5.10 Standard conditions. Unless otherwise specified, the assembly shall meet the performance requirements specified under the following conditions:

Temperature	73 ± 18 degrees Fahrenheit (°F)
Pressure	24.5 - 30.5 inches of mercury (Hg)
Humidity	50 ± 30 percent (%)
Hydraulic fluid temp.	100 ± 10°F
Hydraulic fluid	Per MIL-H-46170

3.6 Environmental conditions. The assembly shall meet the requirements of 3.5 after being subjected to the environmental conditions specified below.

3.6.1 Temperature. The assembly shall be capable of operating at rated flow over the temperature range of -65°F to +275°F, and shall not be damaged when subjected to storage conditions of -65°F. In addition, the assembly shall be capable of withstanding 10 flow cycles of cold start at -65°F. The bubble point shall not be less than the minimum value established during preproduction or initial production testing (see 4.7.3.1).

3.6.2 Vibration. The assembly shall be capable of withstanding the vibration profile per figure 1 for a period of 80 minutes (including up to 4 resonance dwells of 13-1/3 minutes each) in each direction of the 3 mutually perpendicular axes. The filter element media migration shall not exceed 0.004 grams (see 4.7.3.2).

3.6.3 Shock. The assembly shall be capable of withstanding exposure to three half sine wave shock pulses of 40g ± 4g for a duration of 18 ± 3 milliseconds applied in both directions along three mutually perpendicular axes (see 4.7.3.3).

3.7 Identification and marking. Identification and marking shall be in accordance with the requirements of MIL-STD-130, and as specified on Drawings 12257235 or 12270316 (see table VIII).

3.8 Workmanship. The assembly shall be free of defects detrimental to its intended use and service life (see table VIII).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified (see 6.2) in the contract or purchase order, the contractor is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Governemnt for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, Whether indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Inspection equipment. Unless otherwise specified (see 6.2), the supplier is responsible for the provision and maintenance of all inspection and test equipment necessary to assure that supplies and services conform to contract requirements. Commercial, modified commercial, or supplier designated inspection equipment or measuring set-ups must be capable of repetitive measurements to an accuracy of 10 percent of the component tolerance. Calibration of inspection and test equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspections. Classification of inspections shall be as follows:

a. First article (preproduction) inspection (see 4.4).
b. Initial production inspection (see 4.5).
c. Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified, all examinations and tests shall be made under the following conditions:

Temperature	73 ± 18°F
Pressure	24.0 to 30.5 inches of mercury (Hg)
Humidity	50 ± 30% relative humidity
Hydraulic fluid temp.	100 ± 10°F

4.4 First article (preproduction) inspection (see 3.1, 6.3). When specified (see 6.2), a preproduction sample shall consist of four assemblies. The assemblies, properly marked with identifying information, shall be representative of the assemblies proposed to be furnished to the Government. Inspection shall be by methods and procedures specified in table VII.

4.4.1 Failure. Failure of a preproduction assembly to meet the requirements specified herein shall be cause for cessation of inspection. When corrective measures satisfactory to the acquisition activity have been taken, inspection may be continued.

4.5 Initial production inspection (see 3.2, 6.3). Unless otherwise specified (see 6.2), four assemblies from the first ten assemblies produced shall be randomly selected for testing for compliance with the requirements of this specification, by method and procedures specified in table VII.

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TABLE VII. Inspections applied to preproduction and initial production samples.

Title and Sequence	Requirement	Method	Preproduction 1	2	3	4	Initial Production 1	2	3	4
Examinations										
Materials	3.3	Visual	X	X	X		X	X	X	X
Dimensional constraints	3.4.3	Visual	X	X	X		X	X	X	X
Packaging	5.0	Visual						X		
Tests										
Flow direction	3.4.2	4.7.1.1	X	X	X		X	X	X	X
Weight	3.4.4	4.7.1.2	X	X	X		X	X	X	X
Bubble point-alcohol	3.5.1	4.7.2.1.1	X	X	X	X	X	X	X	X
Bubble point - hydraulic fluid	3.5.1	4.7.2.1.2	X	X	X	X	X	X	X	X
Flow rate	3.5.3	4.7.2.3				X				X
Pressure drop	3.5.4	4.7.2.4				X				X
Dirt capacity	3.5.5	4.7.2.5				X				X
Collapse pressure	3.5.7	4.7.2.7				X				X
Flow fatigue	3.5.8	4.7.2.8		X				X		
Filtration	3.5.2	4.7.2.2	X				X			
Efficiency	3.5.6	4.7.2.6	X				X			
Relief valve										
Cracking pressure	3.5.9.1	4.7.2.9.1			X				X	
Flow capacity	3.5.9.2	4.7.2.9.2			X				X	
Reseat pressure	3.5.9.3	4.7.2.9.3			X				X	
Endurance	3.5.9.4	4.7.2.9.4	X	X	X	X	X	X	X	X
Low temperature test	3.6.1	4.7.3.1.1	X	X	X	X	X	X	X	X
Vibration	3.6.2	4.7.3.2	X	X	X	X	X	X	X	X
Shock	3.6.3	4.7.3.3			X				X	

4.5.1 Failure. Failure of an initial production assembly during, or as a result of, initial production inspection shall be cause for rejection of the assembly. The acquisition activity shall refuse acceptance of production assemblies until evidence of corrective action is provided.

4.6 Quality conformance inspection (see 6.3, 6.6).

4.6.1 Lot formation. An inspection lot shall consist of all assemblies of one type, submitted at one time for quality conformance inspection.

4.6.2 Quality conformance examination.

4.6.2.1 Sampling for examination. Samples for examination shall be selected in accordance with inspection level II of MIL-STD-105 from lots specified in 4.6.1.

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4.6.2.2 Classification of defects. For examination purposes, the defects shall be as classified in table VIII.

TABLE VIII. Classification of defects.

Classification	Title	Requirement	Method
Major			
101	Incorrect or faulty materials.	3.3	Visual
102	Dimensions affecting interchangeability not within tolerance.	3.4	SIE 1/
103	Poor workmanship.	3.8	Visual
Minor			
201	Improper packaging.	5.1	Visual
202	Dimensions not affecting interchangeability not within tolerance.	3.4	Visual
203	Incorrect or illegible marking.	3.7	Visual

1/ SIE = Standard Inspection Equipment.

4.6.2.3 Failure. Rejected assemblies and the lots from which they were selected shall be processed in accordance with the acceptance and rejection criteria of MIL-STD-105.

4.6.3 Quality conformance tests.

4.6.3.1 Acceptance test (see 6.6). Each assembly of the inspection lot shall be subjected to test specified in table IX.

TABLE IX. Acceptance test.

Title	Requirement	Method
Bubble point test (production)	3.5.1	4.7.2.1.3

4.6.3.1.1 Failure. Failure of any specimen to pass the tests specified in table VIII shall be rejected.

4.6.3.2 Control tests (see 6.6). Unless otherwise specified, control inspection assemblies shall be selected at the frequencies and subjected to tests specified in table X. These inspections need not be performed on the first control inspection quantity when initial production inspection has been satisfactorily accomplished.

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TABLE X. Control tests.

Title	Requirement	Method	Frequency
Flow rate	3.5.3	4.7.2.3	1/200
Pressure drop	3.5.4	4.7.2.4	1/200
Dirt capacity	3.5.5	4.7.2.5	1/200
Efficiency	3.5.6	4.7.2.6	1/200
Collapse pressure	3.5.7	4.7.2.7	1/200
Filter endurance	3.5.8	4.7.2.8	1/200
Filtration	3.5.2	4.7.2.2	1/200
Relief valve			
Cracking pressure	3.5.9.1	4.7.2.9.1	1/200
Flow capacity	3.5.9.2	4.7.2.9.2	1/200
Reseat pressure	3.5.9.3	4.7.2.9.3	1/200
Endurance	3.5.9.4	4.7.2.9.4	1/200
Temperature	3.6.1	4.7.3.1	1/200
Vibration	3.6.2	4.7.3.2	1/200
Shock	3.6.3	4.7.3.3	1/200

4.6.3.2.1 Control test frequencies. Control testing shall be performed on one (1) assembly at the intervals listed in table IX, except that there shall be not more than one (1) test performed in a two (2) month period nor less than one (1) test in a six (6) month period. The assembly shall be randomly selected from either the first ten (10) assemblies produced for the interval defined, or the last ten (10) assemblies produced from the preceding interval.

4.6.3.2.2 Failure. Failure of a control specimen to meet the specified tests shall be cause for rejection of subsequent assemblies until deficiencies that caused the failure are corrected and proven by testing three (3) consecutively produced assemblies. Control testing shall be limited to the parameters directly related to the failure cause, and parameters affected by corrective action taken.

4.7 Methods of inspection.

a. A filter head assembly suitable for mounting and providing fluid flow through the filter element and relief valve (assembly) shall be provided for testing as required.

b. Unless otherwise specified, the hydraulic fluid used for all tests shall conform to MIL-H-46170 and shall be continuously filtered through a 5 micron absolute filter for the 12257235 assembly and a 3 micron absolute filter for the 12270316 assembly. The contamination level shall not exceed 0.0004 gram per 2000 ml of fluid.

c. Unless otherwise specified, the air used to pressurize the test apparatus shall be filtered through a 0.45 micron membrane filter (type HA Millipore, or equivalent).

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4.7.1 Design characteristic tests.

4.7.1.1 Flow direction verification. Compliance shall be determined in conjunction with 4.7.2.4 through 4.7.2.9 by performing these tests with flow from the outside to the inside (see 3.4.2).

4.7.1.2 Weight. The assembly shall be weighed without hydraulic fluid and shall weigh 1.5 pounds (12257235) or 0.8 pounds (12270316) maximum (see 3.4.4).

4.7.2 Performance tests.

4.7.2.1 Bubble point (see 3.5.1).

4.7.2.1.1 Alcohol, preproduction, and initial production testing. To determine the bubble point for the assembly the following method shall be used:

a. Install the assembly in a test fixture which will permit simultaneous pressurization and 360° rotation of the assembly around its longitudinal axis.

b. Cover the assembly to a depth of 0.5 inch with isopropyl alcohol, or equivalent, at 70° ± 5°F. The bubble point test shall be performed within 10 minutes after immersion of the assembly.

c. Connect an air pressure source to the assembly and raise the pressure in small increments.

d. Rotate the assembly slowly during each pressure increment so that the entire filter surface area can be observed for the appearance of the first bubble.

e. The manometer reading (in inches of water) at the time the first bubble is emitted from the assembly is the alcohol bubble point of the element.

f. Record the data.

4.7.2.1.2 Hydraulic fluid, preproduction and initial production testing, and control level establishment.

a. Install the assembly in a test fixture which will permit simultaneous pressurization and 360° rotation of the assembly around its longitudinal axis.

b. Cover the assembly to a depth of 0.5 inch with test fluid MIL-H-46170, at 70° ± 5°F. The bubble point test shall be performed within 10 minutes after immersion of the assembly.

c. Connect an air pressure source to the assembly and raise the pressure in small increments.

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d. Rotate the assembly slowly during each pressure increment so that the entire filter surface area can be observed for the appearance of the first bubble.

e. The manometer reading (in inches of water) at the time the first bubble is emitted from the assembly is the bubble point of the element.

f. Record the data.

g. The hydraulic fluid bubble point value established per steps (a) through (f) will be the base value used for the remainder of tests required under preproduction and initial production testing.

4.7.2.1.3 Production testing, alcohol. To determine conformance to 3.5.1, the assembly shall be tested as follows:

a. Install the assembly in test fixture which will permit simultaneous pressurization and 360° rotation of the assembly around its longitudinal axis.

b. Cover the assembly to a depth of 0.5 inch with isopropyl alcohol, or equivalent, at 70 ± 5°F.

c. Connect an air pressure source to the assembly and slowly pressurize the assembly to a pressure of 0.1 inch of water less than the minimum alcohol bubble point determined during preproduction testing.

d. Rotate the assembly slowly during pressurization in order to observe the entire filter for bubbles. No bubbles are allowed.

4.7.2.2 Filtration capability (see 3.5.2). The filtration capability shall be determined by subjecting the assembly to the test of 4.7.2.6, steps (a) through (g) with the addition of the following:

a. All hydraulic fluid passed through the test assembly shall be filtered through a 0.8 micron membrane filter disk AA millipore, or equivalent, into a 5000 ml flask. The 5000 ml flask shall then be washed with 1000 ml of prefiltered petroleum ether or prefiltered freon. The cleaning fluid from the washed flask shall also be passed through the membrane filter disk.

b. The entire membrane filter disk shall be scanned and the largest bead present shall be smaller than 5 microns for the 12257235 assembly, and 3 microns for the 12270316 assembly.

4.7.2.3 Flow rate determination. The flow rate for the assembly shall be determined in conjunction with 4.7.2.6 and shall be as specified in table I (see 3.5.3).

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4.7.2.4 Pressure drop test. To determine compliance to the pressure drop across a clean filter element as specified in 3.5.4, the following procedure, or equivalent, shall be used with the relief valve blocked:

Hydraulic pressure, adjusted to provide rated flow, shall be applied to the filter element. Pressure shall be monitored on the inlet and outlet sides of the filter. The pressure drop across the filter element shall be the differences between the inlet and outlet pressure and shall not exceed that specified in table I.

4.7.2.5 Dirt capacity test. To determine conformance to 3.5.5, hydraulic pressure shall be adjusted to provide rated flow in accordance with 3.5.3. AC fine test dust shall be injected into the hydraulic fluid at 4 minute intervals in accordance with table II. The differential pressure across the filter element shall be monitored, and shall not exceed the value specified in table II.

4.7.2.6 Efficiency test. To determine the efficiency of the filter element as specified in 3.5.6, the following procedure, or equivalent, shall be used:

a. Install the test sample into the test setup (see figure 2) with a contamination level of less than 0.0004 gram per 2000 ml of fluid.

b. Temperature of the hydraulic fluid shall be 73 ± 18°F.

c. Add a 5 ml slurry containing contaminants specified in table III into the mixing chamber, and agitate for a minimum of 3 minutes.

d. Apply air pressure to force the fluid containing contaminant through the sample filter assembly and out of the sampling port at rated flow.

e. Collect the filtrate in a 5000 ml flask.

f. Using a suitable wash bottle, wash 750 ml of prefiltered petroleum ether or prefiltered freon through the contaminant mixing chamber and the test filter.

g. Collect the fluid in the same flask as the filtrate.

h. A membrane filter 47-mm diameter disk, absolute 0.8 micron type AA millipore, or equivalent, shall be washed with 200 ml of prefiltered petroleum ether or prefiltered freon, and dried to constant weight at 125°F. The weight shall be accurate to 0.1 milligram. The disk shall be heated at 125°F for 30 minutes and then cooled 30 minutes in a desiccator.

i. The weighed filter membrane shall be assembled into a suitable filter holder assembly and fitted to a 5000 ml vacuum flask connected to a suitable vacuum (minimum 25 inches of mercury (Hg) vacuum).

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j. All collected hydraulic fluid (step g) shall be filtered through the membrane filter. The 5000 ml flask shall then be washed with 1000 ml of prefiltered petroleum ether or prefiltered freon. The cleaning fluids from the washed flask shall also be passed through the membrane filter.

k. Maintaining vacuum, the rim of the membrane filter shall be gently washed with prefiltered petroleum ether or prefiltered freon to remove traces of hydraulic fluid, being very careful not to disturb the cake.

l. The vacuum shall be shut off and the membrane filter removed. It shall be dried to a constant weight at 125°F, and weighed to 0.1 of a milligram.

m. The difference between the membrane filter weights (steps (h) and(l)) shall be determined.

n. Filtration efficiency shall be determined in accordance with the formula specified in 3.5.6 and shall be no less than 97 percent. The difference in weights as determined in step (m) shall be the contaminant value (B), and the contaminant add value (as appropriate) in step (c) shall be the add value (A) as specified in the formula in 3.5.6.

4.7.2.7 Collapse pressure test. To determine conformance to 3.5.7, hydraulic pressure, adjusted to provide rated flow, shall be applied to the filter element. The filter element relief valve shall be blocked. AC Fine Test Dust shall be injected into the hydraulic fluid until the differential pressure of table IV across the filter element is obtained. Subsequent to the test, the filter element shall be inspected for indications of collapse and subjected to the bubble point test of 4.7.2.1.2. The bubble point value obtained shall not be less than the minimum value established during hydraulic fluid preproduction or initial production testing.

4.7.2.8 Flow fatigue test. To determine conformance to 3.5.4 and 3.6.1, the assembly shall be blocked and the temperature of the hydraulic fluid shall be 275°F minimum. The assembly shall be subjected to pressure flow cycles with the fluid flow being increased to a minimum of 30 percent of rated flow, and then shall be decreased to zero flow to complete one cycle. A cycle shall consist of increasing the differential pressure across the assembly from zero to the maximum specified and back to zero by first increasing and then decreasing the flow through the filter element. The filter element shall be progressively loaded with fine dust (AC), or equivalent, to raise the differential pressure (pressure drop) as specified in table V. Peak pressure shall be reached within 25 percent of the cycle and held for at least 50 percent of the cycle. The cycling rate shall be 100 cpm maximum and the number of cycles at each differential pressure shall be in accordance with table V. Upon completion of this test, the filter element shall be bubble point tested in accordance with 4.7.2.1.2. The value obtained shall not be less than the minimum value established during hydraulic fluid preproduction or initial production testing as in 4.7.2.1.2.

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4.7.2.9 Relief valve testing (see 3.5.9).

4.7.2.9.1 Cracking pressure determination. To determine conformance to 3.5.9.1, the filter element media shall be blocked and the assembly installed in a suitable housing. Pressure shall be applied to the inlet port of the filter housing beginning with a pressure of 40 psi and increasing in increments of 20 psi, or less, until cracking pressure of the relief valve in the assembly being tested is reached. At each increment, the pressure shall remain constant for 5 minutes. At each pressure increment, valve opening shall not be considered until the beginning of the third minute. The leakage rate during the last 3 minutes of each pressure increment shall be noted. Cracking pressure shall be considered as the pressure at which the leakage rate through the relief valve amounts to the value specified in table VI.

4.7.2.9.2 Flow capacity determination. To determine conformance to 3.5.9.2, this test shall be conducted in conjunction with 4.7.2.9.1. Pressure shall be increased beyond the cracking pressure, until the flow rate of table VI through the relief valve is obtained. A pressure differential between the inlet and outlet ports shall not exceed the pressure specified in table VI.

4.7.2.9.3 Reseat pressure determination. To determine conformance to 3.5.9.3, this test shall be conducted in conjunction with 4.7.2.9.1. The pressure shall be reduced in 10 psi increments until leakage through the relief valve does not exceed the rate specified in table VI. The resulting pressure shall be considered the relief valve reseating pressure and shall occur at a pressure of not less than that specified in table VI.

4.7.2.9.4 Endurance test. To determine conformance to 3.5.9.4, this test shall be conducted in conjunction with 4.7.2.9.1. The relief valve shall be cycled 2000 times from zero flow to the flow specified in table VI, and back to zero flow. At the conclusion of 2000 cycles, verify that the cracking and reseating requirements of 3.5.9.1 and 3.5.9.3 have been met.

4.7.3 Environmental conditions testing (see 3.6). Assemblies undergoing preproduction, initial production, or control testing shall be subjected to hydraulic fluid bubble point tests (4.7.2.1.2) before and after each of the environmental tests performed. Assemblies undergoing environmental control tests shall have passed the acceptance tests.

4.7.3.1 Temperature (see 3.6.1).

4.7.3.1.1 Low temperature. The assembly shall be subjected to low temperature testing at minus (-) 65°F for 12 hours in accordance with procedure I, method 502 of MIL-STD-810. While stabilized at -65°F, the assembly shall be subjected to a total of 10 flow cycles at a differential pressure across the assembly from zero to 90 psi, and back to zero by first increasing, then decreasing, the flow through the assembly. Each flow cycle shall be 15 ± 5 seconds in duration. The test fluid and assembly shall be

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maintained at -65°F or less for the duration of the test. The flow shall not exceed 150 percent of the rated flow and shall be at a minimum of 0.5 gpm. Subsequent to low pressure exposure, perform the hydraulic fluid bubble point test of 4.7.2.1.2. The bubble point value obtained shall not be less than the minimum value established during preproduction or initial production testing of 4.7.2.1.2.

4.7.3.1.2 High temperature test. This test shall be performed in conjunction with the flow fatigue test of 4.7.2.8.

4.7.3.2 Vibration test (see 3.6.2). The assembly shall be filled with prefiltered hydraulic fluid and the inlet and outlet ports shall be capped with precleaned caps. The assembly shall be tested using procedure VIII, method 514 of MIL-STD-810 with the following conditions.

a. Test level shall be the vibration curve shown in figure 1. Time schedule shall be a total of 80 minutes per axis including 13-1/3 minutes dwell time at each resonance.

b. Connections and instrumentation shall be attached to the assembly to permit cycling as specified in (c) through (f).

c. Prior to vibration test, the housing shall be flushed with 2000 ml of prefiltered hydraulic oil, and effluent contamination level shall be determined. The contaminant value shall represent the blank value "C", as specified in the migration formula below.

d. Conduct resonance search along each axis at reduced vibration amplitude over the frequency range of 5-500 Hz. Up to four (4) resonant frequencies shall be identified (the four most severe resonant frequencies if more than four are present) in each axis for the resonance dwell tests.

e. The assembly unit shall be subjected to resonance dwell and frequency cycling tests with test levels and time schedule as identified in "a" above.

f. Subsequent to vibration tests, the caps shall be removed from the ports, and the assembly shall be flushed with 2000 ml of prefiltered hydraulic oil. The effluent shall be collected and the contamination level shall be determined. The contaminant value shall represent the analysis value "A" as specified in the following migration formula:

$$X = A-C$$

X = Migration analysis value
A = Analysis value
C = Blank value (per step c)

The filter element media migration shall not exceed 0.004 grams.

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g. Perform the hydraulic bubble point test on the flushed assembly in accordance with 4.7.2.1.2. The bubble point value obtained shall not be less than the minimum value established during hydraulic fluid preproduction or initial production testing.

4.7.3.3 Shock (see 3.6.3). The assembly shall be installed in a test fixture and subjected to shock pulses of 40g for a duration of 18 ± 3 milliseconds (ms) per MIL-STD-810, method 514, procedure I. Three (3) half-sine-wave shock pulses shall be applied to the assembly in each direction of 3 mutually perpendicular axes. After the test, the assembly shall be examined for evidence of damage and subjected to the hydraulic fluid bubble point test of 4.7.2.1.2. The bubble point value obtained shall not be less than the minimum value established during hydraulic fluid preproduction or initial production testing as in 4.7.2.1.2.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with the packaging standard for the desired level of protection specified (see 6.2).

5.2 Cleanliness protection compliance. The assembly shall be sealed with a non-contaminant tamper-proof seal to provide the cleanliness protection requirements of MIL-STD-1246, level 200, and shall be adequately tagged and identified before being subjected to preparation for delivery or storage processing.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The filter element and relief valve assembly covered by this specification is intended for use in the hydraulic systems of M60A1, M60A2, and M60A3 tanks operating with hydraulic fluid conforming to MIL-H-46170.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Assembly drawing number.
c. First article inspection, when required.
d. Selection of method and applicable level of preservation and packaging and level of packing of referenced specification (see 5.1).
e. Additional components required to perform control tests (see 4.2.2).

6.3 Test specimen marking. Units subjected to destructive control tests (see 4.7.2 and 4.7.3) are not to be delivered for use as defined in 6.1, and are to be indelibly marked: DO NOT USE.

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6.4 First article preproduction. These inspections verify a vendor's capability to produce a component to a specification and drawing set. The component may be built on a prototype basis without the use of production tooling. They will be normally conducted whenever a new supplier is selected.

6.5 Initial production inspection. These inspections verify the production tooling, methods, and processes used to manufacture a component. They are required on selected first articles produced in a production run. These inspections shall be repeated once every three years on continuing contracts.

6.6 Quality conformance inspection.

6.6.1 Examinations. These inspections verify that physical and configuration characteristics are maintained during the production run.

6.6.2 Control tests. These tests verify that the integrity of the quality level proven by initial production inspection is maintained during the production run.

6.6.3 Acceptance tests. These tests are performed on each manufactured item to verify its functional performance against specification requirements.

6.7 Item definition. The filter element and relief valve assemblies covered by this specification are of the disposable type, 3 and 5 micron filtration, and are suitable for use in the hydraulic control system of combat tanks. The permanent relief valve permits bypassing of the hydraulic fluid in event of excessive restriction to flow through the element.

6.8 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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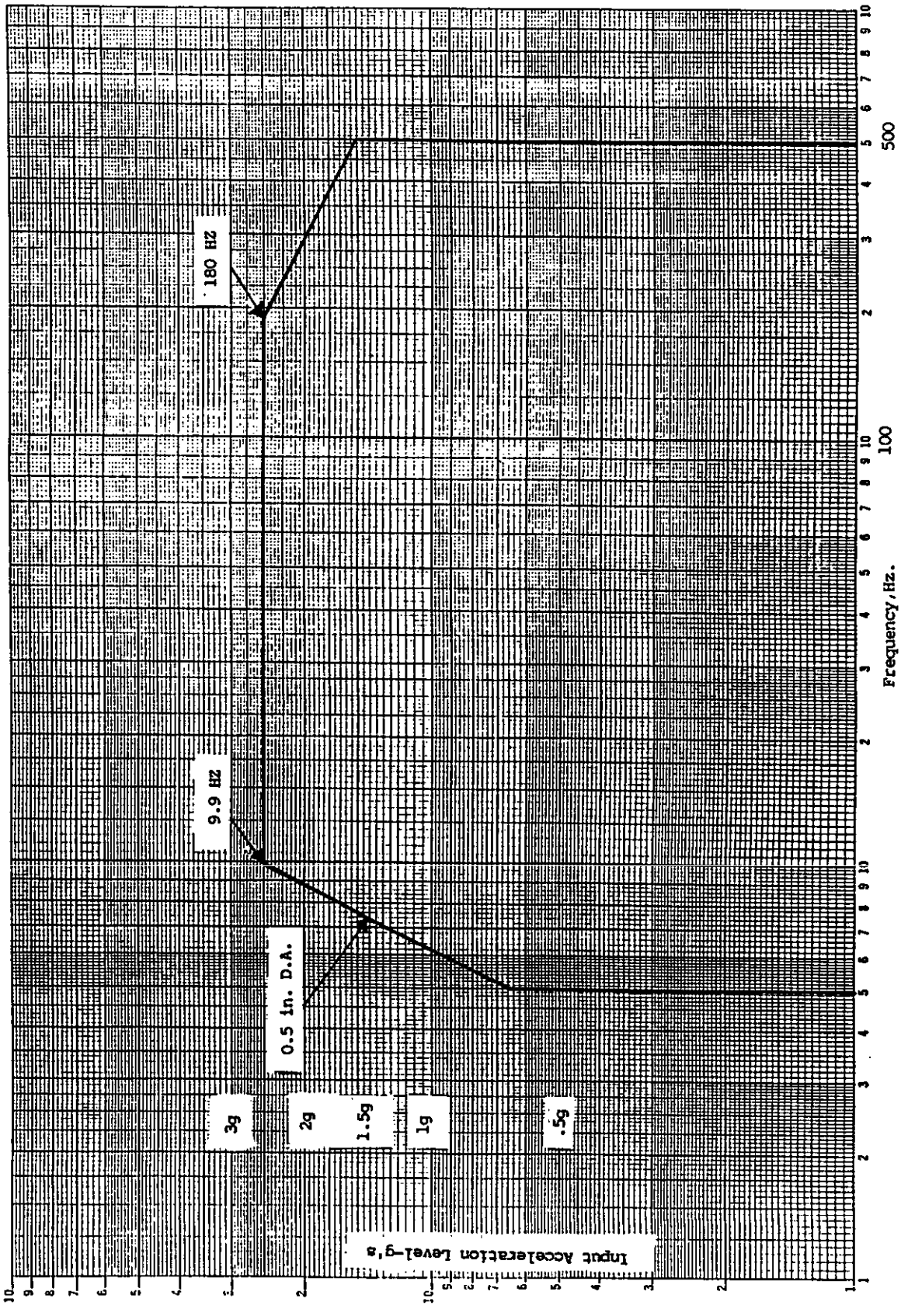


Figure 1. Vibration Profile

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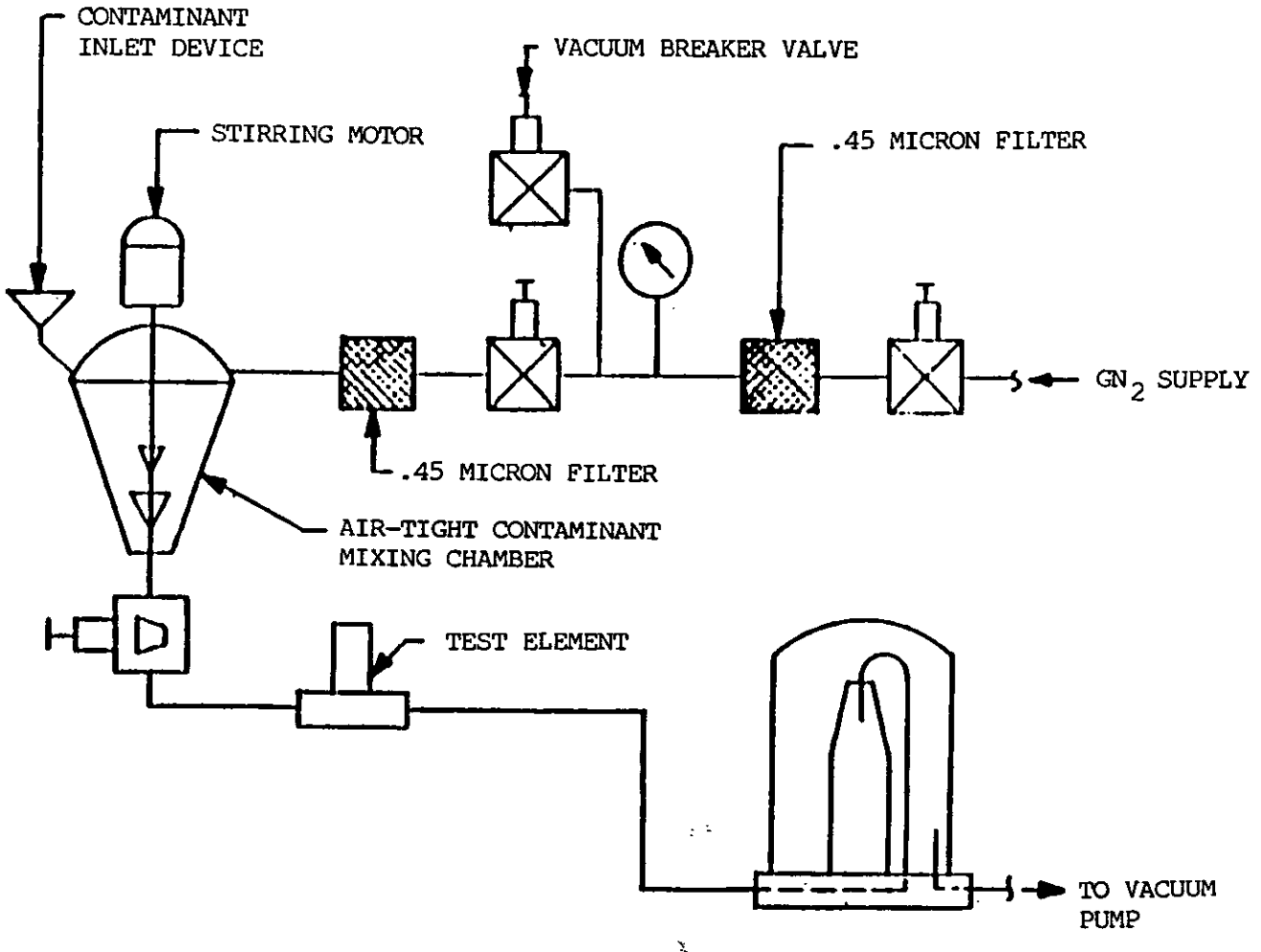


FIGURE 2. Filtration efficiency test set up.

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